GSSSP : Generator of Synthetic Scans of Spectroscopic Plates for Object Detection

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Abstract. GSSSP (Generator of Synthetic Scans of Spectroscopic Plates) is a tool for generating labeled scans of physical spectroscopic observations. The area of spectroscopic analysis is old in the Astronomic Domain. Legacy observational methods recorded stellar spectra on photographic plates. These plates have high historical and scientific value. However, a limited number of them have been digitized, resulting in an insufficient amount of data to train modern computer vision models for observation detection.

In this work, we present a software tool that replicates the appearance of scanned spectroscopic plates by generating synthetic images containing spectra of artificial stellar objects. The generated images are accompanied by label files that enable the training of observation detection models.

We trained a detection model and compared its performance when using only real data, only generated data, and a combination of both. The results show that GSSSP enables the model to learn the general visual patterns of this type of plate, providing a strong generalization basis for subsequent fine-tuning. The resulting model surpasses the current state of the art for this problem.

Keywords: Generated data, Object detection, Spectroscopic plates

1 Spectroscopic Data

Spectroscopy constitutes a fundamental tool for the physical study of astronomical objects. Through the analysis of electromagnetic radiation received, absorbed, or reflected by an object, it is possible to infer several of its physical properties. For example, spectroscopy can reveal the chemical composition, temperature, and radial velocity of astronomical sources. This technique enables the study of celestial bodies beyond their position or apparent brightness in the sky [?].

In the 20th century, the acquisition of spectroscopic data using photographic techniques was widespread. This method consisted of recording spectra through photography, with the distinctive feature that the images were stored on glass plates rather than on photographic paper. Today, many of these plates are preserved in their original institutions [?]. Figure 1 shows a spectroscopic glass plate containing two observations on its surface.

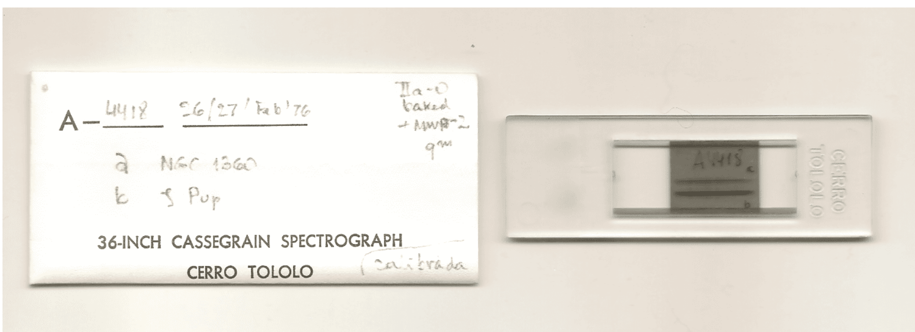
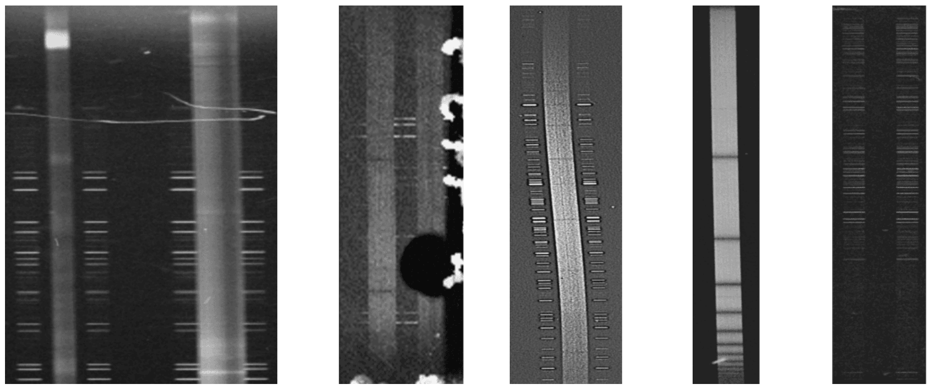


Fig. 1: Right, spectroscopic plate `A4418` of astronomical source `Zeta Puppis` captured at Observatorio Interamericano de Cerro Tololo (Chile) as part of the `ReTrOH` [?] project. The plate contains two spectroscopic observations. Left, wrapping, includes additional information about each observation.

However, storing information in physical plates can present challenges under suboptimal storage conditions. Several incidents affecting this type of heritage have occurred. In 2016, due to a burst pipe, the Harvard College Observatory suffered a flood that affected more than 61,000 plates in its archive, including part of the Harvard plate collection currently digitized by the DASCH project [?]. A similar situation occurred at the Facultad de Ciencias Astronómicas y Geofísicas of the Universidad Nacional de La Plata, where mold was discovered on the furniture used to store the plates [?]. The risk of mold, dust, scratches, or other types of damage is significant and, if not properly addressed, can lead to the permanent degradation of the information stored on each plate.

In addition to physical deterioration, plates may present defects associated with the data acquisition process, such as curvature in the spectral trace, imaging artifacts, or the absence of certain calibration elements. Figure 2 shows examples of plates exhibiting different types of defects.



(a) (b) (c) (d) (e)

Fig. 2: Samples of possible defects on the plates. Defects associated with prolonged storage: (a) Scratches. (b) Dirt and/or moisture stains. Defects associated with the capture process: (c) Curved engraving. (d) Missing comparison lamps. (e) Comparison lamps without their respective spectrum.

To preserve this heritage, institutions such as the International Astronomical Union (IAU) have issued resolutions[5] highlighting the importance of preserving spectroscopic plates for future generations of astronomers. The IAU recommends digitization as a conservation strategy. This process enables the creation of multiple copies and redundant archives, helping to preserve the most relevant information in the long term.

To address these preservation challenges, many institutions have launched digitization projects for their photographic plate collections. Table **??** lists some of the most relevant collections containing large numbers of spectroscopic plates, together with their digitization status and calibration progress.

Plates need to be digitized and calibrated in order to be useful for scientific work. Digitization consists of scanning the plate and storing its information in a digital format. Although the process presents several challenges, different approaches exist to accomplish this, either using adapted scanners [**?**,**?**] or specialized hardware [**?**,**?**,**?**]. On average, scanning a plate takes approximately 15–30 minutes. Calibration involves extracting relevant features from the plate and usually consists of several steps with varying levels of complexity. In previous work, we presented PlateUNLP, a software tool designed to assist in the calibration process. This tool can speed up calibration using machine learning, signal

[5] IAU Resolution B3 (August 2000): Safeguarding the Information in Photographic Observations

Table 1: Some relevant projects or institutions with collections of spectroscopic plates. (a) It is known that HCO has a collection of spectroscopic plates, but the exact number is unknown.

Collection	#plates.	Digitized	Calibrated	Date range	References
`HCO`	?[(a)]	?%	0%	1880–1992	[?]
`APPLAUSE`	19.240	100%	0%	1893–1999	[?]
`ReTrHO`	>15.000	4%	0%	1920–1980	[?]
`WFPDB`	?	100%	?%	1858–2005	[?,?]
`OHP Archive`	63.500	100%	?%	1944–1997	[?]
`Foster Archive`	>4.800	100%	?%	1928–1991	[?]
`DFBS`	1.874	100%	100%	1965–1980	[?]

processing, and metadata-based approaches [?]. In this work, we improve one stage of the calibration process, which consists of detecting observations in plate scans.

2 Detection problem

An observation consists of the spectrum of an astronomical object together with two symmetrical comparison lamps located above and below it. A single plate can contain many observations; however, to extract scientific data, each observation must be isolated in a separate file. Figure 3 shows an example of how the regions corresponding to each observation should be defined. Each observation is defined as a bounding box enclosing the spectrum and its corresponding comparison lamps.

In previous work, we presented a YOLO-based detector [?] that automates observation detection in spectroscopic plates. The model was trained exclusively on real scan images applying data augmentation. The model achieved good performance for this detection task but showed limited precision when predicting the exact boundaries of observations. Although the observations were correctly detected, the predicted bounding boxes were moderately precise.

This behavior can be observed in the IoU-based evaluation metrics: while $mAP_{IoU=0.50} = 0.94$, the performance decreases very to $mAP_{IoU=0.75}$. This indicates that the model can locate the relative positions of the observations but struggles to accurately predict their exact positions and dimensions.

This behavior can be attributed to the small number of scans available for training. As a result, the model lacks sufficient variability in the positions and dimensions of the observations. Data augmentation alleviates this issue; however, because the initial dataset is very small, the generated variations remain limited. Consequently, the model tends to memorize features from the restricted training data rather than generalizing.

In this work, we present a tool to address the problem of limited training data. It enables training with small datasets while maintaining high detection performance for observations with precise positions and dimensions.

Fig. 3: Example of how must be isolated each observation of a spectroscopic plate. The scan of the image contain N observations.

3 GSSSP

GSSSP[6] is a software tool designed to generate synthetic spectroscopic plate scan data. The generated images preserve a realistic appearance and allow the inclusion of artificial defects such as dust stains, moisture marks, reflections, among others. Each generated image is accompanied by labels containing the coordinates and shape of every observation.

The proposed generator constructs synthetic plate scans using a procedural model defined by parameterized numerical operations. This approach allows full control over the generated features and enables the creation of large synthetic datasets even when real data is scarce.

The tool can be installed using package managers such as `pip` or `uv`. The library provides several functions for generating synthetic data directly from code. Users may either call these functions manually or use the implemented generator object, which is compatible with TensorFlow pipelines. The generator internally relies on the same functions, making both approaches equivalent. Figure 4 shows examples of generated plate scans.

The generator provides a high degree of parameterization, allowing control over plate geometry, observation layout, spectral characteristics, and the inclusion of artificial defects.

- Height and width of the canvas.

[6] GSSSP repository

Fig. 4: Generated plate scans with labels of observartions of each plate.

- Background color.
- Inclination of observations within the plate.
- Thickness of each observation component.
- Distance between spectral components.
- Distance between observations.
- Horizontal and vertical deviation of each observation.
- Number of observations per plate.
- Number, intensity, and size of emission lines in each observation component.
- Number, intensity, and size of absorption lines in each observation component.
- Gaussian noise level.
- Horizontal band noise level.
- Number and size of dust stains.
- Blur level.
- Probability of adding a plate edge to the image.
- Number, intensity, and length of horizontal light defects.

All parameters are defined as ranges, ensuring that the generated observations exhibit the desired variability. The produced observations preserve the expected anatomical structure of spectroscopic plates. For example, the comparison lamps located on both sides of each science spectrum are always generated symmetrically.

The generated labels follow the `rel_xywh` format, according to the YOLO annotation scheme [?]. This means that the coordinates of each observation are expressed as normalized values between 0 and 1.

<class_id> <x_center> <y_center> <width> <height>

¡class_id¿ represents the detection class. Since only one class (observations) is considered, its value is always 0. *¡x_center¿* and *¡y_center¿* correspond to the normalized coordinates of the bounding box center, while *¡width¿* and *¡height¿* represent its normalized dimensions. All values except *¡class_id¿* are expressed in the range $[0, 1]$. Figure 4 shows examples of these annotations.

3.1 Available Data

3.2 Available Data

For this study, we use a dataset composed of 361 spectroscopic plates from the archive of the Facultad de Ciencias Astronómicas y Geofísicas of the Universidad Nacional de La Plata. These plates were scanned and annotated for observation detection. The dataset is publicly available[7]. Since the dataset is relatively small, an additional 10,048 synthetic images were generated using GSSSP to complement it.

Each dataset was divided into training and test sets using an 80/20 split:

- Real images (361): 289 for training and 72 for testing.
- Generated images (10,048): 8,038 for training and textcolorred2,010 for testing.

4 Detection model

For model construction, the YOLO architecture was used [**?**]. The model was implemented in TensorFlow using the `keras-cv` library [**?**]. A lightweight version of YOLO, *yolo_v8_s*, was selected for the experiments.

An initial model was trained using only real data in order to establish a baseline performance. This model was then evaluated on both real and generated datasets. If similar performance is obtained on both datasets, this indicates that the generated images exhibit patterns of comparable complexity to the real data.

Additionally, a model was trained using only generated data and evaluated on both real and generated datasets. After, this model was fine-tuned using real data to obtain a final model, which was evaluated on the real dataset.

To evaluate the model performance, Mean Average Precision (mAP) was used. This is a standard evaluation metric for object detection, computed from the precision–recall curve using a defined Intersection over Union (IoU) threshold [**?**,**?**].

IoU measures the overlap between the predicted bounding box and the ground truth bounding box. Its value ranges from 0 (no overlap) to 1 (perfect overlap).

The notation $mAP[IoU = X]$ indicates that the mAP is calculated considering predictions with $IoU \geq 0.X$ as true positives, while predictions with $IoU < 0.X$ are considered incorrect detections. The value of X ranges between 0 and 1. Table 2 shows the metrics obtained in each training experiment.

[7] Observation Detection in Spectroscopic Plates Dataset

Table 2: Results with real, generated and mix data train.

data train	data test	mAP	$mAP[IoU = 0,5]$	$mAP[IoU = 0,75]$
gen-train	*gen-test*			
real-train	*gen-test*			
real-train	*real-test*	0.6597	0.9114	0.6681
gen-train	*real-test*	0.6431	0.8942	0.7675
gen+real-train	*real-test*	0.9360	1.0000	1.0000

5 Conclusions

The results in Table 2 show that the GSSSP tool can generate well-labeled synthetic scans of spectroscopic plates. The generated images capture part of the complexity of real images, and their randomness allows models to be prepared for plates containing many observations with different spectral positions and shapes.

A relevant beheavior is that a small dataset of real scans does not provide sufficient spatial variability in the positions and shapes of observations. This limitation prevents the model from accurately fitting bounding boxes, which negatively affects performance at higher IoU thresholds. For this reason, models trained only on real data and those trained on only generated data achieve similar performance at $mAP[IoU = 0.50]$. However, the model trained on generated data performs better at $mAP[IoU = 0.75]$. This indicates that the synthetic data provides greater variability, allowing the model to predict bounding boxes with higher precision compared to a model trained only on the limited real dataset.

Finally, fine-tuning the model pretrained on generated data with real training data allows the combination of the spatial variability provided by the generator with the complex patterns present in real images. This approach achieves the best overall performance, enabling more accurate and precise detection of observations in real spectroscopic plates. Figure 5 shows example predictions obtained using the fine-tuned model.

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Fig. 5: Predictions obtained with the model trained on generated data and fine-tuned using a limited set of real images.